# Causal Mapping as QDA

 

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# Intro

**21 Sep 2025**

Causal mapping is also a kind of Qualitative Data Analysis (QDQ). How does that even work? This chapter explains.

See also !!Causal mapping is a simple yet powerful form of qualitative coding

Causal Mapping outputs not just codes but a model you can query to answer useful questions

Causal mapping is easy to automate transparently, so is a great fit for scaling with AI

! causal mapping turns QDA on its head

# Causal Mapping outputs not just codes but a model you can query to answer useful questions

> **SOURCE NOTE (consolidation):** This file is a draft/fragment. The flagship QDA-facing paper is now: Causal mapping as causal QDA.
> Companion methods notes: Magnetisation; A simple measure of the goodness of fit of a causal theory to a text corpus.

In the first part of this series !!Causal mapping is a simple yet powerful form of qualitative coding, we argued that causal mapping is a simple yet powerful form of qualitative coding. We showed how its focus on identifying causal links reduces and simplifies the analytical task and provides a clear, structured approach which is relatively easy to apply. However, the true strength of causal mapping lies not just in the coding process itself, but in its output: a **query-able qualitative model**.

Qualitative Data Analysis (QDA) most typically produces a written report alongside a deeper understanding in the minds of the researcher or research team, and hopefully also in the readers. It may also provide other outputs like tables of frequencies or co-occurrences.

We already argued that causal mapping is an especially useful form of QDA because:

- It produces a **structured graph database of causal claims**. This can be viewed as a table or as a map.
- This output isn't a static summary; it's a **dynamic qualitative model** -- a network of interconnected evidence that can be systematically interrogated using standardized, out-of-the-box filters and algorithms.
- The kinds of questions you can answer with a causal map are **particularly useful** because they are about *what causes what -- as seen by your sources*.

## From a List of Themes to a Network of Evidence

Traditional QDA often produces additional outputs like tables of theme frequencies alongside narrative summaries. It is possible to query these to answer questions like "what themes did the younger respondents most often mention when also talking about the main theme".

Causal mapping provides a particularly rich output: a **causal network** -- a qualitative knowledge graph where the factors (which can be understood as themes) are nodes and the causal claims are the links connecting them. This structure is inherently machine-readable and ready for analysis.

Thinking of the output as a **model** is key. Just as a quantitative researcher builds a statistical model to explain relationships in their data, a causal mapper builds a qualitative model of the causal beliefs expressed in texts. This is not unique to our Causal Map app: all applications of causal mapping provide, more or less explicitly, this kind of model, going right back to (Axelrod 1976). This model can then be used to answer new questions, often without needing to go back to the original source texts, though the underlying quotes and context are always available.

## Some standard ways to answer useful questions

Because the output is a structured network, we can apply a range of queries to explore the data. This gives us a library of **pre-existing approaches** to ask **practical questions** about the causal landscape described by the participants.

Here are some questions you can answer using causal mapping.

Individual questions -- introduction ▸

## A corresponding library of filters

The Causal Map app provides about 20 corresponding, ready-to-use filters to answer these kinds of questions, some based on existing causal mapping publications, some new.

In the list of typical queries above:

- Some queries have matching unique pre-defined outputs: a map with a specific filter applied, or a table.
- For some queries, there are different ways to answer them and/or the answer requires more than one filter.
- For a few queries we do not yet have a specific way to answer them in Causal Map.

There are two types of filter:

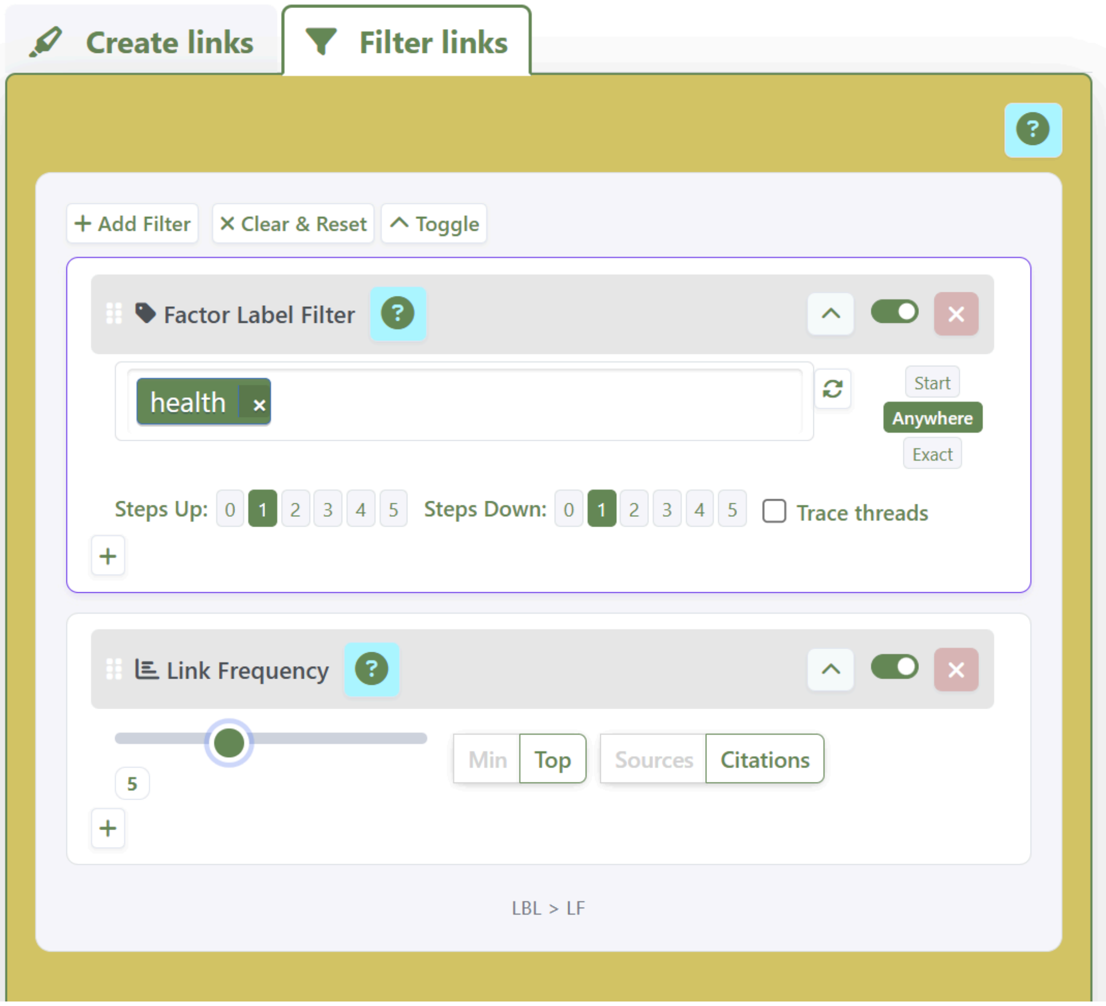
- simple filters which select a subset of the links like links frequency, factor label filter; and path tracing
- "transform filters" like zoom, which temporarily rewrite the cause and/or effect labels.

Here as reference are more details: .

### Chaining filters together

Most importantly **we can chain filters together**, to answer corresponding, composite questions like: *what are the most frequently mentioned upstream influences on these key outcomes, according to the younger respondents?*




Most of the core filters have nothing to do with AI or large language models. They are straightforward, transparent and deterministic. In the Causal Map app, they are available at a click and can be rearranged with drag and drop. But most of them are simple to reconstruct in a spreadsheet or a graph database, without using Causal Map at all.

## Cutting to the chase with causal queries

This ability to systematically query **causal** evidence is what allows causal mapping to **cut to the chase**. For evaluators and researchers, **the core questions are very often about causation**: "Did the program work?" and "How did it work?". Causal mapping structures the evidence provided in the texts to help researchers answer these types of questions.

## A Model of Causal Evidence, Not Causal Facts

It's crucial to be clear about one thing: Causal mapping is **not a method of causal inference**. It does not, on its own, tell you what truly causes what in the real world.

Instead, it creates a qualitative model of **causal claims**. The map organizes what people *said* about causation, allowing the researcher to weigh, compare, and synthesize this evidence. The logic we apply is one of evidence management:

- How much evidence is there for a link between X and Z?
- Is that link direct or indirect?
- Do different subgroups of people agree on this causal pathway?

**Calling these causal claims "evidence" does not mean that anyone should necessarily believe it or that it has been verified[1].** It is simply raw material which we organise and inspect before drawing any conclusions. Researchers can choose to also code additional properties for each link and/or source such as "doubtful" or "reliable" or "verified" and include or exclude links by filtering on those properties.

If the researcher wants to make a causal judgement, they must interpret the map in context, examine the quotes and consider the source of the claims. The map is a powerful tool for structuring and clarifying that judgment.

In any case, **many colleagues use causal mapping not to make causal inferences but simply to understand** *what people think causes what*, and how, for example as a crucial prerequisite for planning policy, communications or interventions.

See also the other caveats we listed in the previous post: [!!Causal mapping is a simple yet powerful form of qualitative coding].

# Conclusion

The real power of causal mapping as a QDA method is that it produces a query-able, qualitative model of causal evidence. This structured output allows researchers and evaluators to apply a range of standardized algorithms to answer practically relevant questions about the causal mechanisms at play, as seen by the sources.

Many researchers and evaluators like using causal mapping to explore their data. In the third part of this series, we will explore how these properties -- a simplified coding task and a structured, query-able output -- make causal mapping especially suited for transparent and verifiable automation with AI.

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1. Thanks to Stève Duchêne https://www.linkedin.com/in/steve-duchene/ for reminding us to clarify this. ↩

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## References


Axelrod (1976). *The Cognitive Mapping Approach to Decision Making*. In *Structure of Decision : The Cognitive Maps of Political Elites*.

# Causal mapping is easy to automate transparently, so is a great fit for scaling with AI

5 Oct 2025

> **SOURCE NOTE (consolidation):** This file is a draft/fragment. The flagship QDA-facing paper is now: Causal mapping as causal QDA.
> Companion methods notes: Magnetisation; A simple measure of the goodness of fit of a causal theory to a text corpus.

## Inter-rater agreement for causal coding is high, making it suitable for automation with AI

The fact that causal coding can be largely reduced to a series of low-level tasks makes it very suitable for automation with AI. High precision and recall scores can be achieved. (Consolidating a large number of in-vivo labels can be accomplished mostly automatically with clustering of text embeddings.)

The AI is used only as a tireless, low-level but incredibly fast assistant with the instruction to code each and every causal claim in the text. This is radically different from the kind of AI-supported “black box” coding which essentially treats the AI as a trusted co-coder who is asked to make, or help make, high-level decisions such as "what are the main themes in this text?" or even “What is the overarching causal network in this text?”.

The accuracy (precision and recall) of AI-supported causal coding is not perfect, but it is improving all the time. Creating, implementing and monitoring the coding protocol remains an essential task ("human in the loop") but we claim that AI-supported causal coding comes closer than other approaches to providing an almost out-of-the-box way to make sense of texts at scale.

The causal coding procedures we have outlined here represent a single-pass, non-iterative approach. Of course this can be expanded to include the more iterative approach essential to most QDA approaches, with varying levels of human oversight. For example we can quickly and cheaply experiment with different coding rules, compare the results, modify the rules and iterate again. This ability to experiment with, compare and iterate potentially hundreds of coding rules and algorithms is a real strength of (semi-)automated coding.

 

Vulnerable to limited attention: if we really process only one section at a time, we will be unable to notice cross-references or places where one section qualifies another, as pointed out by Udo Kelle (1997) xx. This may not be a fundamental limitation of machine-led approaches if we arbitrarily expand the surrounding context, increasing the attention or context window, but at present this is slow and expensive. See A 2023 study by Rezaee et al. compared topic modeling (LDA) vs human qualitative coding of tweets, finding that automated methods reliably find dominant themes but miss subtle frames that human interpretive coding can catch.

 

# Saturation, scale and attention

**9 Nov 2025**

"Reached saturation" mostly means "I've found out about as much on this topic as will fit in any human's head, and my supervisor is happy" doesn't it?
And yes, AIs can in many senses fit more in their heads than humans can. But how many answers do you want?

But just because we can interview 80 thousand people, does that mean we should?

What would we do with all that information? There is a competition for attention.

And actually not just attention in the sense of some kind of narrow spotlight we shine on one thing and then another (out of many), but something like "engagement" -- understanding, making/sharing sense, even caring.

I think many of us have come across the phenomenon where an AI produces a perfectly adequate or even compelling narrative analysis of something but our eyes kind of glaze over and it's hard to care or follow.

This has nothing to do with the ability of the language model, it's got to do with how its products or outputs fit into our world. It wouldn't be any different if the analyses are produced by humans, angels, aliens or LLMs.

Of course it's perfectly possible to give an LLM more information about who we are and what we care about, and ask it to produce results which fit us better, it can do that, up to a point. But we sometimes still struggle to care. Perhaps because we didn't get our hands dirty enough writing the report, or don't have enough skin in the game. That is not a limitation of the LLMs, or of the angels or aliens. It's a by-product of the fact that we can now get almost free, mostly adequate, sometimes even astounding, results to a completely overwhelming range of questions.

[What is the Point of Us. A Sci-Fi Story for Researchers]

 

# Brief review of S Friese – Conversational Analysis to the Power of AI

7 Oct 2025

> **SOURCE NOTE (consolidation):** This is supporting material. The flagship QDA-facing paper is now: Causal mapping as causal QDA.
> Companion methods notes: Magnetisation; A simple measure of the goodness of fit of a causal theory to a text corpus.

This interesting article (Friese 2025) proposes a methodological shift for qualitative data analysis (QDA) that moves beyond traditional coding by introducing **Conversational Analysis with AI (CAAI)**. This approach can be realised by using Dr. Friese's own software, QInsights, replacing the process of coding -- segmenting and labelling data -- with a structured, dialogic interaction between the researcher and a large language model (LLM).

## The Analytical Process

The method uses a five-step process that focuses on synthesis instead of coding.

- **Step 1: Get to know the data.** The researcher uses the AI to create summaries and initial themes to identify key topics for exploration.
- **Step 2: Prepare for analysis.** The researcher picks a topic from the previous step and writes a set of questions to guide the AI conversation. This question list replaces a traditional coding frame and makes the analysis transparent. CAAI analysis proceeds topic by topic. This contrasts with conventional coding frame development.
- **Step 3: Ask questions.** Using the prepared questions, the researcher has a dialogue with the AI about a small subset of the data (e.g., 4-6 interviews). This helps the researcher find patterns and explore surprising findings.
- **Step 4: Synthesize insights.** The researcher slows down, reads the text of the conversation and writes a synthesis of the findings. This can be done alone or collaboratively with the AI. Steps 3 and 4 are repeated for each topic.
- **Step 5: Elevate the analysis (Optional).** The researcher can use the AI to help connect findings to broader theories.

 

## How LLMs are Characterized

The author views Large Language Models (LLMs) not as intelligent beings but as useful analytical partners. Their value comes from being trained on vast amounts of human-written, "socially-situated corpora". The article states that models lack true human understanding or lived experience, a concept referred to as "Seinsverbundenheit". However, their outputs are still insightful because they reflect the patterns and "lifeworlds" present in their training data. The AI's knowledge is described as a "collective and distributed" echo of human meaning-making. In this role, the LLM acts as not only in a deductive and inductive fashion but also as an "abductive catalyst"—a tool that surfaces unexpected connections and provokes new ideas for the researcher, without needing to be intelligent itself.

This approach reframes the researcher's role from a coder to a conductor of an analytic conversation, prioritizing interpretation and synthesis over mechanical categorization. Following (Krähnke et al. 2025), Friese characterises this process as hermeneutical -- an approach in which meaning emerges through the researcher engaging in an open interaction with a text, in this case with the addition of a third party, the AI, which provokes and questions the process. "Meaning does not reside in the data itself but is generated by a recursive relationship between the analyst, the context, and the text" (p. 6).

Rigor and trustworthiness are established not through coding frames or inter-coder agreement, but through the transparency of the documented dialogue, traceability to source data via retrieval-augmented generation (RAG) systems (provision of quotes), and the somewhat replicable nature of the question sets that guide the inquiry. Ultimately, CAAI presents a post-coding paradigm where analysis emerges directly from a dynamic and reflexive engagement with the data, mediated by AI.

In the course of the article, Friese revisits the text analysis she conducted for her own PhD and candidly concludes that she could have done better, and much faster, using the method she proposes.

# Strengths

## Reflection on use of LLMs

Firstly, I'd congratulate Dr. Friese for presenting a **deeply argued explanation of and justification** for the QInsights workflow. As researchers, however we use LLMs, it is crucial that we continue to reflect on what we are doing, rather than letting the muscles of our critical thought go weak through lack of exercise.

 

## LLM-supported QDA as hermeneutics

Secondly, the advent of generative AI is a very good time to ask again: do we really need coding to make QDA rigorous? As she says: "qualitative research is entering a **moment of methodological experimentation**". After all, rigorous coding alone was never enough to make text analysis rigorous, not least because coding can ignore context and the positionality of the researcher and the research task. **Re-positioning LLM-supported QDA in the broader context of hermeneutics** is useful and refreshing.

## Dangers of consumer-facing LLMs

Thirdly, she reminds researchers that **methodological rigour is very hard to achieve with unreflective use of consumer-facing LLMs** like ChatGPT, not least because they will skim-read the text to find a quick answer and will too often grab a possibly incorrect answer from their training data rather than ground it in the actual text. What's more, a platform like QInsights can provide a completely documented workflow, where every conversation and decision made is recorded -- although we have no insight into the AI's 'thought processes' behind each of its contributions to this workflow beyond what it tells us.

## A new kind of epistemic actor between human and machine

I usually find arguments about whether generative AIs are "really" conscious or can "really" understand something as simply spurious, and no more interesting than arguing about whether a computer can *really* "copy" or "save" or "read" or even "predict" something. However, Friese has a really interesting angle on this, an angle which takes us further rather than trying to police useful language. She '**treats AI as a new kind of epistemic actor** — neither a mere stochastic parrot nor a conscious knower, but as something else: a dialogic partner capable of generating insight through probabilistic modelling trained on socially situated corpora. “Construction” in CAAI is not solely human; it is distributed, emerging through interaction between human interpretation, data context, and machine-generated associations.' So an AI is more than "an assistant": it is an assistant which possesses, or *is*, a whole world of interconnected meanings.

# Some caveats

Against those very notable strengths, here are a couple of caveats.

## How new is this? If it is new, is that really because of the inclusion of AI?

So, we can drop coding and use an AI as a sort of meta-assistant to co-create meaning from text. My biggest question is: to the extent that the AI is not that different from a tireless, lightning-fast and knowledgeable human assistant, couldn't we drop the AI from the methodological equation and just say, here's one newish way to analyse texts hermeneutically? (Though you'd have to have plenty of time and either one very good human assistant or a team of normal ones for this to actually work.)

Is this new method really just a speeded-up version of what we could in theory have done with very fast human assistants? Or is does the sheer magnitude of that speed-up mean a kind of Hegelian transformation of quantity into quality: it's just so much faster that the result is qualitatively different? Or is there something else about the method which is fundamentally new?

## Skipping the coding step does lose some reproducibility -- does the AI really change this?

If the addition of AI support does not make this way of working fundamentally new, I don't really see how it gives us a free pass on all the original reasons for using coding in the first place. This method is presumably, if everything else is held equal, less reproducible than a method which does employ coding.

Dr. Friese does mention "Reapplication of refined questions across subgroups; independent synthesis by multiple researchers; replication over time" as possible ways to *assess* reproducibility.

This is where our approach to causal coding at [Causal Map] differs most strongly from CAAI: we only use the AI as a low-level assistant with narrowly defined tasks, so the workflow is less dependent on the stochastic nature of the AI's behaviour and even confines human input to the most crucial high-level decisions, and so is easier to reproduce.

## Why this particular set of steps?

It isn't hard at least in theory to think of endless variations of the 4 or 5 steps set out in the article -- for example the team-based approach suggested by (Krähnke et al. 2025), so what are the specific arguments for this particular set of steps? Why for example do we work topic by topic? I'm sure it's not the case that "anything goes", but why not?

## Whose world?

I am sure Dr. Friese would acknowledge that the world of "machine-generated associations" underpinning AI responses is not just given but is constructed in a very specific way by very specific organisations on a very specific set of training data. This fact has been repeated almost to exhaustion in recent writings on AI, but she does elevate this world to a kind of new meta-assistant, so it would be good to reflect once more on its make-up and provenance. You could see the meta-assistant as a combination of all human lifeworlds, except that of course it isn't --

 

because many people, and certain continents, and whole swathes of actual human non-digital life are not equally included.

## In conclusion

As Dr. Friese says, generative AI potentially opens up a whole world of possibilities for qualitative text analysis and indeed for social science in general. We need to be acutely aware of the multiple social, political, methodological and environmental risks of this technology, and *at the same time* not miss out on its benefits.

And, try out QInsights!

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### References


Friese (2025). *Conversational Analysis with AI - CA to the Power of AI: Rethinking Coding in Qualitative Analysis*. https://doi.org/10.2139/ssrn.5232579.

Krähnke, Pehl, & Dresing (2025). *Hybride Interpretation textbasierter Daten mit dialogisch integrierten LLMs: Zur Nutzung generativer KI in der qualitativen Forschung*. DEU. https://www.ssoar.info/ssoar/handle/document/99389.


 

# GenAI is distracted by shiny things

**7 Oct 2025**

I just read another great post from Susanne Friese. I love the way she is forging different ways to combine reflexive qualitative research and GenAI. GenAI surely opens up many new possibilities for qualitative research, including using it for coding and without coding.

But I don't really buy the main argument. Here's why.

My worry with conversational approaches can be centred on this sentence:

> "This shift becomes tangible when researchers conduct a theme analysis and see that major categories—previously developed through weeks of coding—reappear as coherent, evidence-linked themes within minutes."

Braun & Clarke rightfully criticise human researchers for saying "the themes emerged from the text". But this is just the same, isn't it?

"What are the major themes in this document" is often presented as easy, low-level tasks suitable for an AI. But they are not trivial. They are fundamentally creative and meaning-making and involve a whole mass of evaluative judgements. What on earth is a theme? For whom?

Susanne continues:

> "Researchers are repositioned. Their central task becomes judgment: asking meaningful questions, evaluating plausibility, integrating context, articulating theory, and remaining reflexively aware of technological mediation."

But you already handed a mass of judgement-making to your AI when you asked it to identify themes. And in particular Susanne does this right at the start of the workflow, in step 1 (Friese 2025).

Coding is presented as a necessary evil, some kind of crutch on the way to sensemaking which we can now throw away. But where does coding come from? Long before GenAI, a researcher could immerse themselves in a set of texts and eventually emerge with any number of declarative, findings illustrated by quotes. Were they too distracted by the first shiny thing they saw in the text? Who knows? Did they pay too much attention to some parts of it and ignore other uncomfortable parts? In order for qualitative analysis to count as any kind of science, different frameworks and guidelines were constructed to ensure that at least some parts of the analysis were more systematic, retraceable (nachvollziehbar) and transparent (and perhaps even reproducible). There are endless ways to do that, to show that the research process was not distracted by the first shiny thing it saw and ignored the uncomfortable parts. A human, who is essentially distractible by shiny things, can delegate parts of the process to "mere coding" in order to partially mitigate that risk. But **we cannot mitigate humans' propensity to be distracted by shiny things by delegating tasks to a GenAI which is at least as distractible by shiny things**. I know that conversational approaches **do** bring in other methods and procedures to mitigate these risks, having read Friese (2025), but I think this could be better spelled out in Susanne's post.

I agree with Susanne that GenAI may be bringing about a paradigm shift in qualitative research, also in its relationship to quantitative research. But that paradigm shift can take many forms. The way we use GenAI to scale causal mapping is another really different way. Causal mapping happens to be based much more on coding. We would argue that it is therefore more systematic than one which starts by asking "what are the themes here?"

But there are also surely **hundreds** of other ways to use GenAI in qualitative research, most yet to be discovered.

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## References

Friese (2025). *Conversational Analysis with AI - CA to the Power of AI: Rethinking Coding in Qualitative Analysis*. https://doi.org/10.2139/ssrn.5232579.

 